

## Module 5

# Valuing Bonds and Stocks

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FINE 3010, Financial Management  
Prof. Renping Li, Tulane University

# From Module 4 to Module 5

- **Module 4**, the time value of money, the present value of future cash flows

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- **Module 4**, the time value of money, the present value of future cash flows
- **Module 5**, the same time-value-of-money principles value bonds and stocks
  - Bond, the present value of its coupons and face value
  - Stock, the present value of its expected dividends

# Every Financial Asset Is Debt or Equity

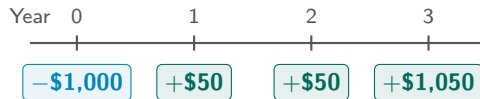
- **Financial asset**, value from a contractual right or ownership claim



# Debt's Cash Flow Is Fixed, a Stock's Is Not

## Debt

\$1,000, 3 years, 5% interest

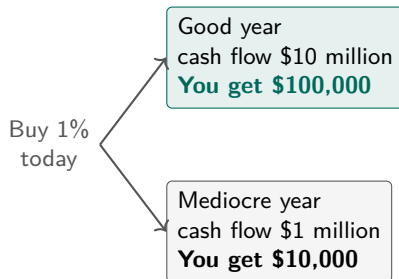


Every payment known in advance

## Stock

1% of the company

Next year



Value of claim depends on how the company does

# Common Types of Corporate Financing

**Debt**, lending to the company

Main forms, bonds and loans

Holder gets a fixed promised amount

“Senior”, paid back before equity in  
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No voting rights or control

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**Equity**, stock ownership in the company

Holder gets dividends

“Residual claimant”, paid whatever is left after all promised payments, no guaranteed payout

Voting rights

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## Module 5, Part 1

# Bonds

# Bonds, Definitions

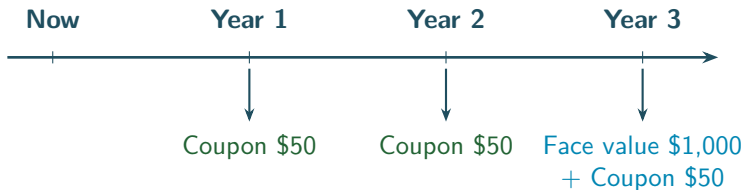
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# Bonds, Definitions

- **Bond**, a security that obligates the issuer (the borrower) to make specified payments to the investor (the lender, or “bondholder”)
- Typically some periodic payments (the “coupon”), then a “face value” at the end
- E.g., a three-year bond with an annual coupon of \$50 and a face value of \$1,000, the borrower promises to pay,



# Bonds, Terminology

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- **Coupon rate**,

$$\text{Coupon rate} = \frac{\text{Annual coupon}}{\text{Face value}}$$

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  - E.g., **Module 4**'s Puerto Rico bond, the coupon was 0 but the interest rate was 5.15%

# Example, Puerto Rico Borrows

Recall from **Module 4**

- In 2007, Puerto Rico sold bonds that promised to pay \$1,000 after 47 years (no periodic “coupon” payments)
- These bonds were sold at \$94.40 each
- What was the interest rate paid by Puerto Rico?



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Recall from **Module 4**



$$N = 47, \quad I = ?$$

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  - 0 **PMT**
  - 1000 **FV**
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- The rate is 5.15%

# Bonds, Pricing

- **Bond value**, the present value of all its cash flows (coupons and face value), discounted at the “required rate of return”  $r$ 
  - “Required rate of return”, the name for the discount rate in this application

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- E.g., for a bond with time to maturity  $T$ ,

$$\text{Value} = \text{PV} = \frac{\text{Coupon}_1}{(1+r)^1} + \frac{\text{Coupon}_2}{(1+r)^2} + \dots + \frac{\text{Coupon}_T + \text{Face value}}{(1+r)^T}$$

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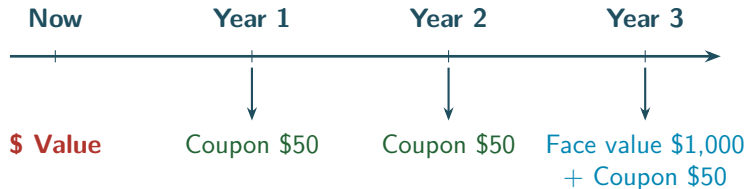
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- Investors choose the opportunity cost, “what could I get investing in something else with a similar risk”, as the required rate of return  $r$

## Example, Bond Pricing

- A bond has an annual coupon of \$50 and a face value of \$1,000, with three years to maturity



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$$\text{Value} = \frac{\$50}{(1 + 0.04)^1} + \frac{\$50}{(1 + 0.04)^2} + \frac{\$50 + \$1,000}{(1 + 0.04)^3} = \$1,027.75$$

## Example, Bond Pricing with a Financial Calculator

- The same bond, an annual coupon of \$50, a face value of \$1,000, three years to maturity, and a required rate of return of 4%
- Use a financial calculator to value the bond

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  - **CPT**, then **PV**

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- Use a financial calculator to value the bond
- Solution,
  - 1000 **FV**
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  - 3 **N**
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  - **CPT**, then **PV**
- The output is  $-\$1,027.75$  (why the minus sign? recall how financial calculators get to the answer in [Module 4](#))

# Valuing Bonds with a Financial Calculator

- The financial calculator keys map onto a bond,
  - **N**, the number of periods (years) until maturity
  - **PV**, the value of the bond (the present value of the coupons and the face value)
  - **PMT**, the coupons (in dollars)
  - **FV**, the face value (the lump sum paid at the end)
  - **I/Y**, the discount rate (the required rate of return  $r$ )

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- Companies issue bonds (and other debt) to raise financing from investors
- In return, the companies make pre-specified payments of coupons and face value to the investors
- The value of a bond is the discounted payments, at the required rate of return

# Bond Yields

- **Yield to maturity (YTM)**, the discount rate  $y$  for which the present value of the bond's payments (coupons and face value) equals its price (also simply the “yield”)

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- **Yield to maturity (YTM)**, the discount rate  $y$  for which the present value of the bond's payments (coupons and face value) equals its price (also simply the "yield")
- Given the price of a bond (PV), its maturity  $T$ , and its coupons, find the yield to maturity by solving for  $y$ ,

$$\text{Price} = \frac{\text{Coupon}}{(1+y)^1} + \frac{\text{Coupon}}{(1+y)^2} + \cdots + \frac{\text{Coupon} + \text{Face value}}{(1+y)^T}$$

## Example, Bond Yields

- Solving for the YTM analytically is hard, so use the financial calculator
- A bond's annual coupon rate is 5%. Its face value is \$1,000. Its maturity is 3 years. Its current market price is \$1,027.75. What is its YTM?

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  - **CPT**, then **I/Y**
- The yield is 4%

## Practice Problem

- The table shows data for three bonds. Each bond's coupon is zero. Each bond's face value is \$1,000

| Bond | Price | Maturity (years) | Yield to maturity |
|------|-------|------------------|-------------------|
| A    | \$310 | 25               | ?                 |
| B    | \$310 | ?                | 9%                |
| C    | ?     | 11               | 8%                |

- What is the yield to maturity of bond A?
- What is the maturity of bond B?
- What is the price of bond C?

## Practice Problem

- Solution, on the financial calculator (the highlighted key is the one solved for),

|           | <b>N</b> | <b>I/Y</b> | <b>PV</b> | <b>PMT</b> | <b>FV</b> |
|-----------|----------|------------|-----------|------------|-----------|
| a. Bond A | 25       | ?          | −310      | 0          | 1,000     |
| b. Bond B | ?        | 9          | −310      | 0          | 1,000     |
| c. Bond C | 11       | 8          | ?         | 0          | −1,000    |

## Practice Problem

- Solution, on the financial calculator (the highlighted key is the one solved for),

|           | N  | I/Y   | PV   | PMT | FV     |
|-----------|----|-------|------|-----|--------|
| a. Bond A | 25 | 4.796 | -310 | 0   | 1,000  |
| b. Bond B | ?  | 9     | -310 | 0   | 1,000  |
| c. Bond C | 11 | 8     | ?    | 0   | -1,000 |

- a. The yield to maturity of bond A is 4.796%

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- a. The yield to maturity of bond A is 4.796%
- b. The maturity of bond B is 13.59 years

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| c. Bond C | 11    | 8     | 428.88 | 0   | -1,000 |

- a. The yield to maturity of bond A is 4.796%
- b. The maturity of bond B is 13.59 years
- c. The price of bond C is \$428.88

## Practice Problem

- A firm's bonds pay an annual coupon rate of 6.8%. Their face value is \$1,000. Their price is \$795.69. Their yield to maturity is 9.50%
- What is the remaining maturity of these bonds?

## Practice Problem

- Solution, the bond price is the PV of the coupon payments plus the PV of the face value,

Bond price = PV of coupon payments + PV of face value

Bond price =  $C \times ((1 \div r) - \{1 \div [r(1 + r)^t]\}) + FV \div (1 + r)^t$

$\$795.69 = (0.068 \times \$1,000) \times ((1 \div 0.095) - \{1 \div [0.095 \times (1 + 0.095)^t]\}) + \$1,000 \div (1.095)^t$

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  - 9.5 **I/Y**, -795.69 **PV**, 68 **PMT**, 1000 **FV**, then **CPT** **N**

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- On the financial calculator,
  - 9.5 **I/Y**, -795.69 **PV**, 68 **PMT**, 1000 **FV**, then **CPT** **N**
- The remaining maturity is about 14 years ( $N = 13.98$ )

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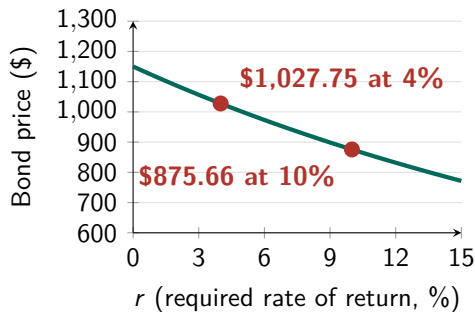
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  2. Risk of default (the opportunity cost for risk-taking)
    - Higher (systematic) risk of default, higher  $r$

# Interest Rates and Bond Prices Move in Opposite Directions

- Because the bond price is,

$$\text{Price} = \frac{\text{Coupon}_1}{(1+r)^1} + \dots + \frac{\text{Coupon}_T + \text{Face value}}{(1+r)^T}$$

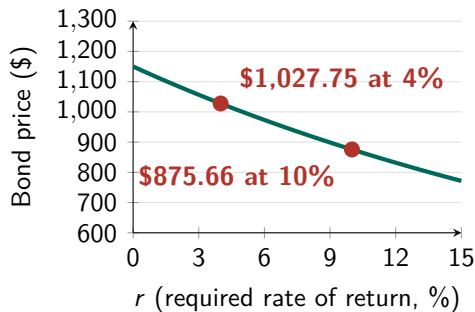


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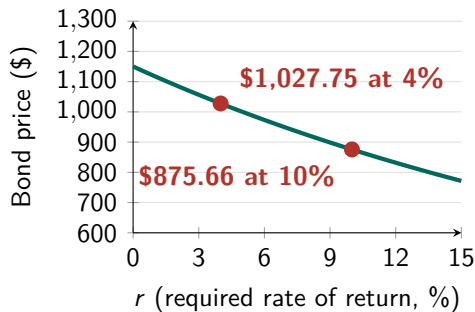


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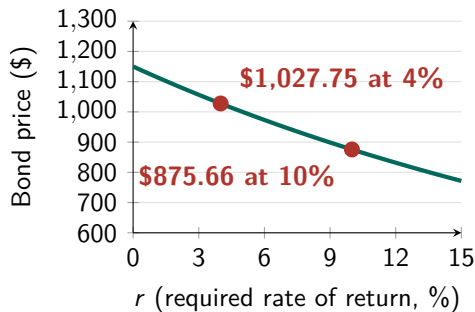


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- **Default premium**, the additional yield on a bond that investors require for bearing credit risk

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- **Investment-grade bonds**, rated Baa or higher by Moody's, or BBB or higher by Standard & Poor's
- **High-yield bonds** ("junk" bonds), rated Ba or BB and lower

# Default Risk, Credit Ratings

| Moody's                       | S&P     | Default within 10 yr |
|-------------------------------|---------|----------------------|
| <b>Investment-grade bonds</b> |         |                      |
| Aaa                           | AAA     | 0.1%                 |
| Aa                            | AA      | 0.7%                 |
| A                             | A       | 2.2%                 |
| Baa                           | BBB     | 3.5%                 |
| <b>High-yield bonds</b>       |         |                      |
| Ba                            | BB      | 15.7%                |
| B                             | B       | 35.5%                |
| Caa, Ca                       | CCC, CC | 48.9%                |
| C                             | C       | n/a                  |

Source, Brealey, Myers, and Marcus, Table 6.2

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| Aa                            | AA      | 0.7%                 |
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- Yields tend to be higher (prices lower) for lower-rated bonds

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- Interest rates and bond prices move in opposite directions
- $r$  rises with prevailing interest rates and with default risk, lower-rated bonds carry higher yields

## Module 5, Part 2

# Stocks

# Goals

- **Part 1, bonds**
  - Differences between debt and equity
  - Valuation
  - Yields
  - Interest rates
  - Default risk
- **Part 2, stocks**
  - Valuation, dividend discount model
  - Valuation, comparables
  - Market efficiency

# Valuing Stocks

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  1. Dividend discount model
  2. Valuation by comparables
- The approaches are complementary, each has advantages and disadvantages, and no approach provides “the” right answer
- Professionals use a combination
- You will learn more in “FINE 4100 Advanced Financial Management” and “FINE 4110 Investments in Equities”

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- **Dividend yield**,  $\frac{\text{Dividend}}{\text{Stock price}}$
- **Earnings per share (EPS)**,  $\frac{\text{Net income}}{\text{Number of shares outstanding}}$
- **P/E ratio**,  $\frac{\text{Stock price}}{\text{Earnings per share}}$

# Valuing Stocks, Dividend Discount Model

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- What price should you pay for the stock today ( $P_0$ )?

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$$P_0 = \frac{\text{Div}_1}{(1 + r_e)^1} + \frac{\text{Div}_2}{(1 + r_e)^2} + \frac{\text{Div}_3}{(1 + r_e)^3} + \cdots + \frac{\text{Div}_\infty}{(1 + r_e)^\infty} = \text{PV}(\text{Dividends})$$

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- **Required return on equity** ( $r_e$ ), the discount rate, set by the opportunity cost of investing in something with similar risk

# Valuing Stocks, Dividend Discount Model

- Suppose you plan to buy a stock and own it forever
- What price should you pay for the stock today ( $P_0$ )?
- Stock value, the present value of the cash flows you get from owning it,

$$P_0 = \frac{\text{Div}_1}{(1 + r_e)^1} + \frac{\text{Div}_2}{(1 + r_e)^2} + \frac{\text{Div}_3}{(1 + r_e)^3} + \cdots + \frac{\text{Div}_\infty}{(1 + r_e)^\infty} = \text{PV}(\text{Dividends})$$

- **Required return on equity** ( $r_e$ ), the discount rate, set by the opportunity cost of investing in something with similar risk
- $\text{Div}_1, \text{Div}_2, \dots$  are *expected* dividends

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- The latter formula is often called the “**Gordon growth model**”

## Example, Gordon Growth Model

- 1. A firm will pay a \$3.00 dividend forever (the first dividend exactly a year from now). Its required return on equity is 12%. What will investors pay for the stock?
- 2. What if the firm's stock pays a \$3.00 dividend a year from now, and then the dividend grows at 8% forever?

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- That is, dividend yield plus expected capital appreciation (in %)

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  - If the firm is only temporarily not paying dividends, the dividend discount model still applies
  - But future dividends are then probably even more uncertain!

# Valuing Dividends with Non-Constant Growth

- Estimate the dividends year by year over a horizon of  $H$  years, then assume constant growth after year  $H$ ,

$$PV = \frac{\text{Div}_1}{(1 + r_e)^1} + \frac{\text{Div}_2}{(1 + r_e)^2} + \dots + \frac{\text{Div}_H}{(1 + r_e)^H} + \frac{P_H}{(1 + r_e)^H}$$

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- You will learn more in “FINE 4100 Advanced Financial Management” and “FINE 4110 Investments in Equities”

## Example, Valuing Non-Constant Growth

- A stock's expected dividends are in the table. Its required return on equity is 8.5%. Its dividends grow at 6% a year beginning in year 6. What is the value of the stock?

| Year     | 1      | 2      | 3      | 4      | 5      |
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$$PV_{\text{Stock}} = 10.23 + 89.11 = \$99.33$$

## Practice Problem

- Consider the following three stocks,
  - a. Stock A is expected to pay a dividend of \$10 a share forever
  - b. Stock B is expected to pay a dividend of \$5 next year. Thereafter, dividend growth is expected to be 4% a year forever
  - c. Stock C is expected to pay a dividend of \$5 next year. Thereafter, dividend growth is expected to be 20% a year for 5 years (years 2 through 6) and zero thereafter
- Each stock's required return on equity is 10%. Which stock is the most valuable? What if the required return on equity is 7%?

# Practice Problem

- Solution, at a 10% required return on equity,

10% capitalization rate:

$$P_A = \text{DIV}_1 / r = \$10 / 0.1 = \$100$$

$$P_B = \text{DIV}_1 / (r - g) = \$5 / (0.1 - 0.04) = \$83.33$$

$$\begin{aligned} P_C &= \frac{\text{DIV}_1}{1+r} + \frac{\text{DIV}_1 \times (1+g)}{(1+r)^2} + \frac{\text{DIV}_1 \times (1+g)^2}{(1+r)^3} + \frac{\text{DIV}_1 \times (1+g)^3}{(1+r)^4} + \frac{\text{DIV}_1 \times (1+g)^4}{(1+r)^5} + \\ &\quad \frac{\text{DIV}_1 \times (1+g)^5}{(1+r)^6} + \frac{(\text{DIV}_1 \times [1+g]^5 \times [1+g_2]) / r}{(1+r)^6} \\ P_C &= \$5 / 1.1 + (\$5 \times 1.2) / 1.1^2 + (\$5 \times 1.2^2) / 1.1^3 + (\$5 \times 1.2^3) / 1.1^4 \\ &\quad + (\$5 \times 1.2^4) / 1.1^5 + (\$5 \times 1.2^5) / 1.1^6 + ([\$5 \times 1.2^5 \times \{1 + 0\}] / 0.1) / 1.1^6 \\ P_C &= \$104.51 \end{aligned}$$

At a 10% capitalization rate, Stock C has the largest present value.

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At a 10% capitalization rate, Stock C has the largest present value.

- At a 7% required return on equity,  $P_A = \$142.86$ ,  $P_B = \$166.67$ , and  $P_C = \$156.50$ , so Stock B is the most valuable

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- E.g., a 2,000 sqft house in New Orleans sells for \$400,000. What is your estimate for a neighboring house of 2,500 sqft?

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- Underlying idea, we can “scale” prices!

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- 4. Back out Company X’s implied value (the numerator) as the average multiple times Company X’s denominator

$$P_X = (\text{Average P/E of comparables}) \times \text{EPS}_X$$

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- Solution, multiply Herman Miller's EPS by the P/E of comparable firms,  
$$P = (\text{Average P/E of comparables}) \times \text{EPS} = 21.3 \times 1.38 = \$29.39$$

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- The question asks for the share price, not the market cap,

$$P = \frac{\text{Market cap}}{\text{Number of shares outstanding}} = \frac{36.8 \text{ billion}}{1.2 \text{ billion}} = \$30.67$$

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- Assumes the comparables are priced correctly
  - Cannot tell if an entire industry is overvalued or undervalued
  - E.g., no help in valuing tech firms during the late-1990s “dot-com bubble”

# Buying and Selling Stock on the Stock Market

- Current stock price

## FedEx Corporation (FDX)

☆ Add to watchlist

NYSE - Nasdaq Real Time Price. Currency in USD

**239.98** -9.04 (-3.63%)

As of 1:02PM EDT. Market open.

Summary

Chart

Conversations

Statistics

Profile

Final

|                |                 |                          |                             |
|----------------|-----------------|--------------------------|-----------------------------|
| Previous Close | 249.02          | Market Cap               | 64.318B                     |
| Open           | 246.00          | Beta                     | 1.57                        |
| Bid            | 239.98 x 100    | PE Ratio (TTM)           | 22.11                       |
| Ask            | 240.10 x 100    | EPS (TTM)                | 10.86                       |
| Day's Range    | 236.00 - 246.00 | Earnings Date            | Mar 19, 2018 - Mar 23, 2018 |
| 52 Week Range  | 182.89 - 274.66 | Forward Dividend & Yield | 2.00 (0.80%)                |
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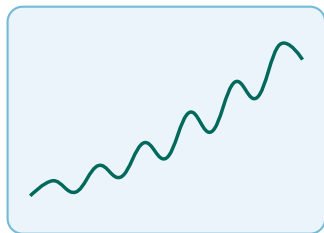
- Market capitalization
- **Beta**, a measure of risk (how much the stock moves with the overall market)
- Price/earnings
- EPS
- Dividend and dividend yield

# Can You Beat the Stock Market?

- It's really hard!

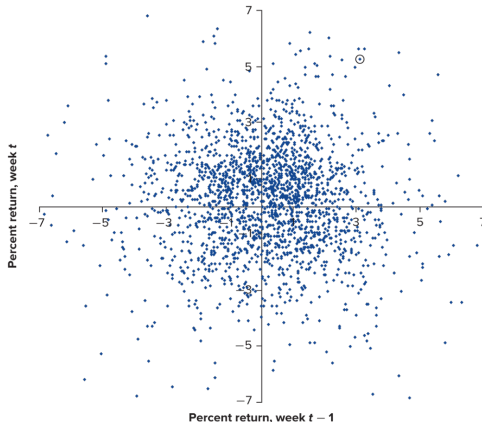
# Can You Beat the Stock Market?

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- **Technical analysis**, investors who attempt to forecast stock prices by watching the fluctuations in historical prices (“wiggle watchers”)



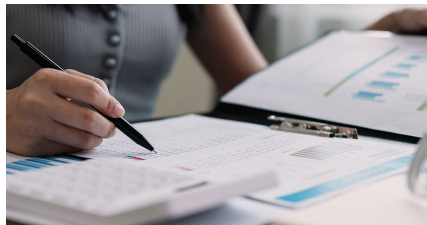
# Discuss, Where Is the Pattern?

- Scatter plot of the NYSE Composite Index return in one week against its return in the previous week



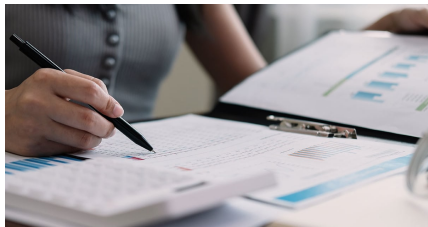
# Can You Beat the Stock Market?

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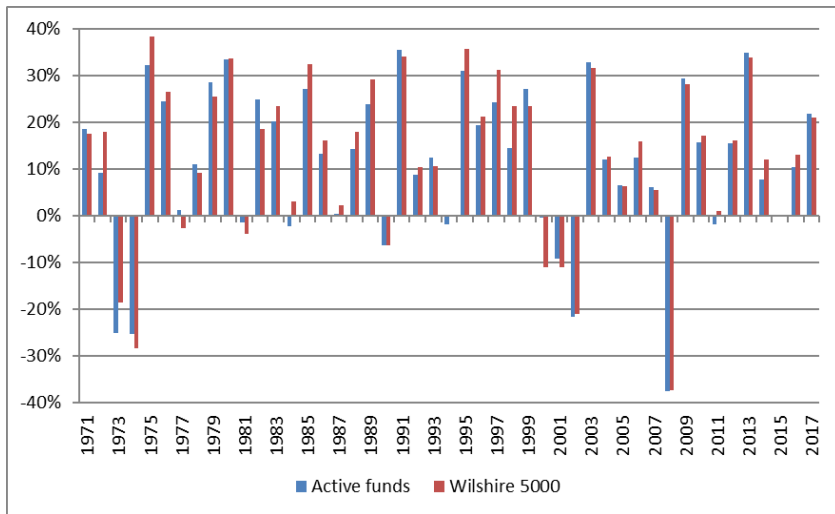


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- **Fundamental analysis**, investors who attempt to find mispriced securities by analyzing fundamental information, such as accounting data and business prospects
- Also really hard!



# Professional Fund Managers Don't Generally Beat the Index!



# Information in Stock Prices

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- **Efficient markets hypothesis**, “securities are fairly priced, given all information that is available to investors”
- What **type** of information is captured in the stock price?
  1. All historical price information? (“weak” efficiency)
  2. All public, easily available information? (“semi-strong” efficiency)
  3. Also all private or difficult-to-interpret information? (“strong” efficiency)

# What Does This Mean for Investors?

## Pros 👍

- If stocks are fairly priced, even an investor who **knows nothing** can buy **any** stock and expect it to fairly compensate them for the risk they take

## Cons 👎

- To *consistently* make money above the *fair* (*risk-compensating*) return by betting on publicly traded shares, an investor needs a very distinct competitive advantage
- E.g., expertise or access to information that only a few people can figure out

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- Comparables, scale a peer multiple (P/E, market-to-book) to the firm being valued
- Beating the market is really hard, prices already reflect available information (the efficient markets hypothesis)

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- Interest rates and bond prices move in opposite directions
- $r$  rises with prevailing interest rates and with default risk, lower-rated bonds carry higher yields
- Stocks, the dividend discount model or comparables, and efficient markets make the index hard to beat

# Thank you!

Please let me know if you have any questions.